SQL Questions for assignment

1.Find the 2nd highest salaried person in the finance department and also find his name?

```sql
-- 1. Find the 2nd highest salaried person in the finance department and also find his name

SELECT e.first_name, e.last_name, s.salary  -- select employee name and salary
FROM employees e
JOIN salaries s ON e.emp_no = s.emp_no -- join employees with salaries
WHERE e.emp_no IN (
    SELECT emp_no
    FROM dept_emp
    WHERE dept_no = (
        SELECT dept_no
        FROM  departments
        WHERE dept_name = 'Finance' -- select finance department
    )
)
ORDER BY s.salary desc  -- sort salaries from highest to lowest
LIMIT 1 OFFSET 1; -- get 2nd highest salary
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

first_name	last_name	salary
Jiang	Vesel	134379

2.Fetch the list of of employees who worked in Production department between 1998 and 2000?

```
-- 2.Fetch the list of of employees who worked in Production department between 1998 and 2000

SELECT e.emp_no, e.first_name, e.last_name
FROM employees e
JOIN dept_emp de ON e.emp_no = de.emp_no        -- join employees with departments
WHERE de.dept_no = (
    SELECT dept_no
    FROM departments
    WHERE dept_name = "Production"              -- get Production department id
)
AND de.from_date BETWEEN "1998-01-01" AND "2000-12-31";  -- who worked between 1998 and 2000
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

emp_no	first_name	last_name
10024	Suzette	Pettey
10096	Jayson	Mandell
10113	Jaewon	Syrzycki
10131	Magdalena	Eldridge
10162	Florina	Eugenio
10192	Mohua	Falck
10284	Masali	Murrill
10307	Nikolaos	Leaver
10317	Arie	Birge
10337	Jeong	Sadowsky
10344	Uwe	Garnier
10382	Shirish	Dredge
10430	Ferdinand	Chenney
10529	Ymte	Yetto
10552	Kazuhito	Shiratori
10593	Falguni	Kenevan
10598	Margo	Gronowski

3.Who is the 3rd highest salaried person in Production department without using orderby.

```
-- 3.Who is the 3rd highest salaried person in Production department without using orderby
SELECT e.emp_no, e.first_name, e.last_name, s.salary
FROM employees e
JOIN salaries s ON e.emp_no = s.emp_no
WHERE s.salary = (
  SELECT MAX(salary) FROM salaries
  WHERE emp_no IN (SELECT emp_no FROM dept_emp
                   WHERE dept_no=(SELECT dept_no FROM departments WHERE dept_name="Production"))
  AND salary < (
    SELECT MAX(salary) FROM salaries
    WHERE emp_no IN (SELECT emp_no FROM dept_emp
                     WHERE dept_no=(SELECT dept_no FROM departments WHERE dept_name="Production"))
    AND salary < (
      SELECT MAX(salary) FROM salaries
      WHERE emp_no IN (SELECT emp_no FROM dept_emp
                       WHERE dept_no=(SELECT dept_no FROM departments WHERE dept_name='Production'))
    )
  )
);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_no	first_name	last_name	salary
61036	Shimshon	Nergos	129361

4.Find how many employees joined in which year in a complete single report.

```
-- 4 .Find how many employees joined in which year in a complete single report

SELECT YEAR(hire_date) AS joined_year,
        COUNT(emp_no) AS total_employees -- total_emp total no.of employees
FROM employees
GROUP BY YEAR(hire_date) -- Group by each year
ORDER BY joined_year; -- ascending order
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

joined_year	total_employees
1985	35316
1986	36150
1987	33501
1988	31436
1989	28394
1990	25610
1991	22568
1992	20402
1993	17772
1994	14835
1995	12115
1996	9574
1997	6669
1998	4155
1999	1514
2000	13

5.Fetch the maximum employees joined year from the above report without using having.

```
-- 5.Fetch the maximum employees joined year from the above report without using having

SELECT YEAR(hire_date) AS joined_year
FROM employees
GROUP BY YEAR(hire_date)
ORDER BY COUNT(emp_no) DESC -- Maximum employees joined year
LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

joined_year
1986

6.Need the reports total cost divided yearly in a table of all departments.

```
-- 6.Need the reports total cost divided yearly in a table of all departments.

SELECT d.dept_name,
       YEAR(s.from_date) AS year,            -- year of salary
       SUM(s.salary) AS total_cost           -- total salary cost
FROM departments d
JOIN dept_emp de ON d.dept_no = de.dept_no   -- de joins employees to departments
JOIN salaries s ON de.emp_no = s.emp_no      -- get salary of each employee
GROUP BY d.dept_name, YEAR(s.from_date)      -- group by department and year
ORDER BY d.dept_name, year;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dept_name	year	total_cost
Customer Service	1985	21622993
Customer Service	1986	47369131
Customer Service	1987	73010083
Customer Service	1988	101352441
Customer Service	1989	130030259
Customer Service	1990	159320515
Customer Service	1991	189543590
Customer Service	1992	221596131
Customer Service	1993	251110669
Customer Service	1994	282421834
Customer Service	1995	312254799
Customer Service	1996	344411025
Customer Service	1997	374708287
Customer Service	1998	404098974
Customer Service	1999	431073691
Customer Service	2000	437843946
Customer Service	2001	437424703
Customer Service	2002	254269912
Development	1985	86021286

7.Find the Year where highest female employees joined.

```
-- 7.Find the Year where highest female employees joined.
SELECT YEAR(hire_date) AS join_date
FROM employees
WHERE gender ="f" -- female employees
GROUP BY YEAR(hire_date)
ORDER BY COUNT(emp_no) DESC
LIMIT 1; -- highest female employees
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

join_date
1986

8.Hire Date of the current manager of Human Resources department.(hint: to_Date year is 9999)

```sql
-- 8.Hire Date of the current manager of Human Resources department

SELECT e.hire_date
FROM employees e
JOIN dept_manager dm ON e.emp_no = dm.emp_no
WHERE dm.dept_no = (
    SELECT dept_no
    FROM departments
    WHERE dept_name = 'Human Resources'
)
AND YEAR(dm.to_date) = 9999;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

hire_date
1985-08-04

9.Salary of the current manager of the Finance dept

```sql
-- 9.Salary of the current manager of the Finance dept

SELECT s.salary
FROM salaries s
JOIN dept_manager dm ON s.emp_no = dm.emp_no
WHERE dm.dept_no = (
    SELECT dept_no
    FROM departments
    WHERE dept_name = 'Finance'
)
AND YEAR(dm.to_date) = 9999      -- current manager
AND YEAR(s.to_date) = 9999;      -- current salary
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

salary
83457

10.Current dept managers first names along with emp_no.

```
-- 10.Current dept managers first names along with emp_no

SELECT e.emp_no, e. first_name
FROM employees e
JOIN dept_manager dm ON e.emp_no = dm.emp_no
WHERE YEAR(dm.to_date) = 9999;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_no	first_name
110039	Vishwani
110114	Isamu
110228	Karsten
110420	Oscar
110567	Leon
110854	Dung
111133	Hauke
111534	Hilary
111939	Yuchang